

- 7 (a) Define *electric field strength* at a point.

The electric field strength at a point is defined as the electric force exerted per unit positive charge acting on a test charge placed at that point.

.....[1]

- (b) Electrons are emitted from a cathode C and are accelerated towards an anode A, as illustrated in Fig. 7.1.

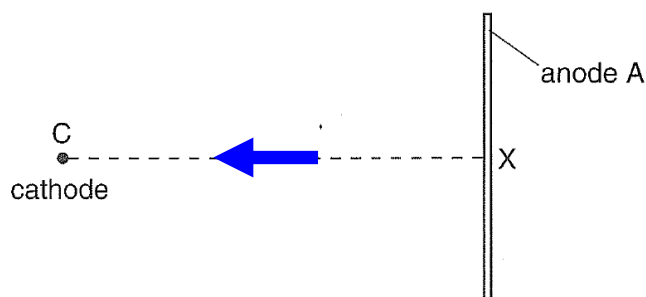
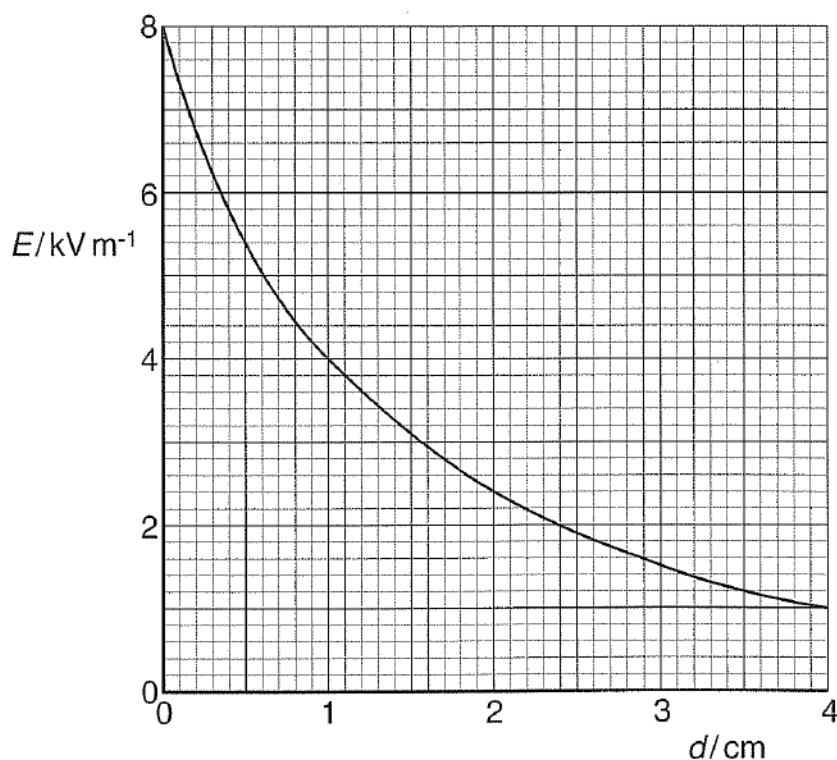


Fig. 7.1

The anode is earthed. CX is a line drawn from C normal to the anode A. The distance CX is 4.0 cm.

The variation with distance d from C along CX of the magnitude of the electric field strength E is shown in Fig. 7.2.



- (i) On Fig. 7.1, mark with an arrow the direction of the electric field along CX. [1]

- (ii) Use Fig. 7.2 to determine the force F on an electron at a point mid-way between C and X.

At $d = 2 \text{ cm}$, $E = 2.4 \text{ kV m}^{-1}$ [C1]

$$F = qE$$

$$= (1.60 \times 10^{-19})(2.4 \times 10^3)$$

$$= 3.84 \times 10^{-16} \text{ N} \text{ [A1]}$$

$$F = \dots\dots\dots \text{ N [2]}$$

- (c) (i) A student assumes that the force F on the electron remains constant as the electron moves from C to X.

Use the value of F calculated in (b)(ii) to estimate, on the basis of this assumption, the potential difference between C and X.

$$\text{Work done on charge} = Fd$$

$$q\Delta V = Fd$$

$$\Delta V = \frac{Fd}{q}$$

$$= \frac{(3.84 \times 10^{-16})(4.0 \times 10^{-2})}{1.60 \times 10^{-19}} \text{ [M1: must use (b)(ii) answer]}$$

$$= 96 \text{ V [A1]}$$

$$\text{potential difference} = \dots\dots\dots \text{ V [2]}$$

- (ii) Suggest, with a reason, whether the magnitude of the potential difference calculated in (i) will be an over-estimate or an under-estimate of the actual potential difference.

Under-estimate because (by observation) the area under graph is
greater than 96 V or answer in (c)(i).

.....

.....[1]

[Total: 7]

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Section B

Answer **one** question from this Section in the spaces provided.